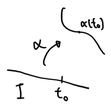
$\alpha: I \to \mathbb{R}^3$, $t_0 \in I$

define $s: I \to \mathbb{R}$

$t \rightsquigarrow \int_{t_0}^{t} |\alpha'(u)|\,du = \int_{t_0}^{t} (\alpha)$



The function $I \to \mathbb{R}$, $u \rightsquigarrow |\alpha'(u)|$ is in general just continuous.

s is just C^1. In particular, s is not diffeom

$s'(t) = |\alpha'(t)|$

If $|\alpha'(t)| \neq 0$ then s is indeed a diffeom between I and $s(I)$

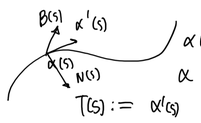
Such curves are called regular $\Rightarrow s \to C^\infty$

call $J = s(I)$ $\quad I \underset{\varphi}{\overset{s}{\rightleftarrows}} J$ φ is the inverse of s

and define $\beta(s) = \alpha \circ \varphi : J \to \mathbb{R}^3$

$\beta'(s) = \alpha'(\varphi(s))\ \varphi'(s) = \dfrac{\alpha'(\varphi(s))}{|\alpha'(\varphi(s))|}$

$|\beta'(s)| = 1 \quad \forall s \in J$



α is p.a.l.

$T(s) := \alpha'(s) \qquad |T(s)| = 1 \quad \forall s$

$\langle T(s), T(s)\rangle = 1 \Rightarrow$ (taking deriv)

$\langle T'(s), T(s)\rangle + \langle T(s), T'(s)\rangle = 0$

$\langle T'(s), T(s)\rangle = 0$

T' is always orthogonal to T.

Def: 1) $k(s) = |T'(s)|$ is called curvature of α at s

This is only continuous. It is C^∞ if $T'(s) \neq 0$ $\forall s$

From now on we assume $k(s) > 0$ $\forall s$

2) $N(s) = \dfrac{T'(s)}{|T'(s)|} = \dfrac{T'(s)}{k(s)}$ is called the normal vector to α at s

3) $B(s) := T(s) \wedge N(s)$ is called the binormal vector to α at s

4) $\{T(s), N(s), B(s)\}$ is called Frenet's thirdedron.

Def: $\alpha: I \to \mathbb{R}^3$ is parametrized with arc length (p.a.l.) if $|\alpha'(t)| = 1 \quad \forall t$.

Theorem: Every regular curve can be parametrized with one length

[Notes 1) s is not unique it depends on t_0 and and on a constant of integration

Ex: 1) Lines $\alpha(t) = t\underline{v} + \underline{v_0}$, $t \in \mathbb{R}$ $\quad t_0 = 0$

$\alpha'(t) = \underline{v}$ $\quad (\underline{v} \neq 0)$ $\quad s(t) = \int_0^t |\underline{v}|\,du = t|\underline{v}|$

$\varphi(s) = \dfrac{s}{|\underline{v}|}$ $\quad \beta(s) = \dfrac{s}{|\underline{v}|}\cdot \underline{v} + \underline{v_0}$

2) $\alpha: \mathbb{R} \to \mathbb{R}^2$ Logarithmic spiral

$t \rightsquigarrow (ae^{bt}\cos t,\ ae^{bt}\sin t)$, $a>0$, $b<0$, $t_0 = 0$

$s(t) = \int_0^t |\alpha'(t)|\,dt$

$= \int_0^t \left|\left(ae^{bt}(b\cos t - \sin t),\ ae^{bt}(b\sin t + \cos t)\right)\right| dt$

$= \int_0^t ae^{bt}\left(b^2\cos^2 t + \sin^2 t + b^2\sin^2 t + \cos^2 t\right)^{1/2} dt$

$= \int_0^t a e^{bt}(b^2+1)^{\frac{1}{2}}\,dt = \dfrac{e^{bt}-1}{b}\cdot a(b^2+1)^{\frac{1}{2}}$

$T'(s) = k(s)N(s)$

$B(s) = T(s) \wedge N(s)$

$B'(s) = T'(s)\wedge N(s) + T(s)\wedge N'(s) = T(s)\wedge N'(s)$

$\langle B, B\rangle = 1$

$\langle B', B\rangle = 0$

$\begin{cases} \langle B', T\rangle = 0 \\ \langle B', B\rangle = 0 \end{cases}$

$B'(s) = \tau(s)N(s)$

$\tau(s)$ is called torsion of α at s

$\tau = \langle B', N\rangle \Rightarrow$ smooth

$T' \rightsquigarrow k$

$B' \rightsquigarrow \tau$

$N' \rightsquigarrow ?$

$\langle N, T\rangle = 0 \Rightarrow \langle N', T\rangle + \langle N, T'\rangle = 0$

$\langle N, B\rangle = 0 \Rightarrow \langle N', B\rangle + \langle N, B'\rangle = 0$

$\langle N', T\rangle + k(s) = 0$

$\langle N, B\rangle + \tau(s) = 0$

Theorem (Frenet's Formula) $\alpha: I \to \mathbb{R}^3$ p.a.l. and $k > 0$

$\begin{cases} T' = kN \\ N' = -kT - \tau B \\ B' = \tau N \end{cases}$

Ex 1) Line: $\alpha(t) = t\underline{v} + \underline{v_0}$ $\quad |\underline{v}| = 1$

$T(t) = \underline{v}$ $\quad k(t) = 0$

$T' = 0$

2) Circle: $\alpha(t) = r(\cos\frac{t}{r}, \sin\frac{t}{r}, 0) + \underline{c}$

$\alpha'(t) = r\left(-\frac{1}{r}\sin\frac{t}{r}, \frac{1}{r}\cos\frac{t}{r}, 0\right)$

$= \left(-\sin\frac{t}{r}, \cos\frac{t}{r}, 0\right)$

$\Rightarrow t$ = a.l. $\quad T = \alpha'$

$T' = \left(-\frac{1}{r}\cos\frac{t}{r}, -\frac{1}{r}\sin\frac{t}{r}, 0\right) \Rightarrow k(t) = \frac{1}{r}\ (\neq 0)$

$N = \left(-\cos\frac{t}{r}, -\sin\frac{t}{r}, 0\right)$ $\quad B(t) = (0,0,1) \Rightarrow B' = 0$

$\Rightarrow \tau = 0$

Theorem: $\alpha: I \to \mathbb{R}^3$, p.a.l., $k > 0$

α is planar iff $\tau \equiv 0$

Pf. Suppose $\tau \equiv 0$: $B(s) = \underline{v}$ (constant vector) $|\underline{v}| = 1$

$\left(\langle \alpha(s), \underline{v}\rangle\right)' = \langle T, \underline{v}\rangle = 0$

$\Rightarrow \langle \alpha(s), \underline{v}\rangle = a$, $a \in \mathbb{R}$ which is the eq. of a plane in $\mathbb{R}^3$

Theorem: $\alpha: I \to \mathbb{R}^3$ p.a.l.

$k \equiv 0$ iff α is a piece of a line.

Pf $\leftarrow$: done

$\rightarrow$ $k \equiv 0 \Rightarrow T(s) = \underline{v}$ constant vector

$\alpha' = T$

$\alpha = t\underline{v} + \underline{v_0}$